

Yoshi Sato

Undergraduate Researcher | Computer Security

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EDUCATION

WASEDA UNIVERSITY

Bachelor of Science in Computer Science and Communications Engineering

Tokyo, Japan

Oct 2022 - Jul 2026

Research Interests: Microarchitectural Security, Systems Security, Trusted Execution Environments (TEEs)

PUBLICATIONS

[Conference Paper] **Y. Sato**, H. Uranami, A. Saiki, K. Kimura. "Towards GPU Passthrough in Intel TDX: Design Challenges and Early Baselines." In *Proceedings of the IEEE 23rd International Conference on Dependable, Autonomic and Secure Computing (DASC)*, 2025. (Best Poster Award)

[Submitted / Under Review] M. Barbaraci, S. Ma, H. Malvai, M. Mouallem, S. Ren, **Y. Sato**, S. Yang, F. Zhang. "DeadDrop: Responsible Disclosure of Smart Contract Bugs." Submitted to *Proceedings on Privacy Enhancing Technologies (PoPETS)*, 2026. (Author order alphabetical)

RESEARCH EXPERIENCE

Research Intern, Decentralized Systems Group, Yale University

Jul 2025 - Present

Advisor: Prof. Fan Zhang

- Architected the core cryptographic primitive for DeadDrop using Oblivious Message Retrieval (OMR) to produce the first functional prototype for oblivious bug reporting.
- Identified a critical "database jumbling" vulnerability where implementation deviated from the theoretical binding of payloads and clues; led discussions to redefine the system's threat model.
- Designed and executed performance evaluations to quantify clue processing, achieving a throughput of 32,768 payloads in 108s (single-core) and scaling to 524,288 payloads in 110s (16-core).
- Parallelized the precomputation phase of OMR to **reduce initialization latency by 65%**, significantly improving the prototype's practicality for real-world deployment.
- Leading a Systematization of Knowledge (SoK) on TEE project with researchers from Yale and UIUC to define a new taxonomy and evaluation framework for confidential containers.

Student Researcher, Kasahara-Kimura Lab, Waseda University

Apr 2025 - Present

Advisor: Prof. Keiji Kimura & Prof. Hironori Kasahara

- Initiated an independent research project on GPU passthrough in Intel TDX, culminating in a paper accepted to IEEE DASC 2025 that won the Best Poster Award.
- Dissected microarchitectural I/O bottlenecks by comparing shared memory buffers in AMD SEV-SNP and Intel TDX.
- Identified that while AMD's PVALIDATE incurs zero VM Exits, Intel TDX's TDG.MEM.PAGE.ACCEPT triggers mandatory VM Exits, creating a structural latency bottleneck.
- Quantified memory overheads, establishing a baseline revealing that pageable memory throughput degrades by ~53% compared to pinned memory in confidential environments.

Research Assistant, IIJ Research Laboratory

Mar 2025 - Present

Advisor: Dr. Pierre-Louis Aublin

- Collaborated with the LSDS Group at Imperial College London on Serverless Confidential Containers (SC2) to analyze performance overheads in secure serverless computing.
- Transformed a research prototype into a fully reproducible testbed, enabling the first performance evaluation of the SC2 system on Intel TDX hardware.
- Engineered custom network proxy support for the Knative control plane to resolve critical deployment failures in restricted network environments.
- Analyzing confidential VM cold-start execution paths using Ftrace to identify and quantify dominant latency sources in the boot process.

AWARDS

Best Poster Award, IEEE DASC 2025

PROFESSIONAL EXPERIENCE

IoT Penetration Testing Intern, FFRI Security Jan - Feb 2025

- Conducted static and dynamic binary analysis on IoT firmware and malware executables using Ghidra and GDB, identifying critical vulnerabilities in legacy software components.
- Executed penetration tests on simulated IoT environments to simulate real-world attack vectors against embedded devices, using Nmap, Metasploit, and Wireshark.
- Presented vulnerability findings and remediation strategies to senior engineers, translating complex technical results for cross-functional stakeholders.

LEADERSHIP

Secretary, IEEE-HKN, Mu Tau Chapter Nov 2025 - Present

Nominated by 2018 IEEE Computer Society President and invited to chapter leadership calls and discussions. Formal induction upcoming later this year.

Student Volunteer Jun 2025

International Symposium on Computer Architecture (ISCA) 2025, Tokyo, Japan

Co-founder, Kuma Lab Apr 2024 - Present

- Founded a student research community of ~30 members to foster peer learning in AI and robotics.
- Organized a flagship technical workshop hosted at Google Shibuya for 25 attendees to bridge the gap between advanced research and non-STEM audiences.
- Delivered a technical lecture on Gaussian Splatting for 3D reconstruction, while coordinating member demonstrations on robot teleoperation and CV classification.

SKILLS

- TEEs: Intel TDX, AMD SEV-SNP
- Systems: QEMU, Kubernetes, Knative, Docker, Archlinux
- Applied Cryptography: Microsoft SEAL, PALISADE
- Programming Languages: Python, C++, Rust
- Cybersecurity: Ghidra, GDB, Wireshark